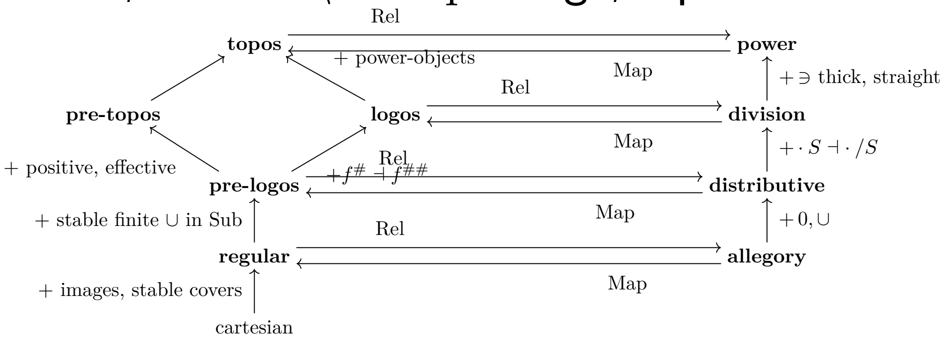
Categories & Allegories — the taxonomy

the two towers, the Rel $\rightleftarrows$ Map bridge, representation theorems



Rightward: $\mathrm{Rel}(C)$ is always unitary tabular. Leftward: for A **unitary tabular**, each right-hand property gives $\mathrm{Map}(A)$ the matching left-hand one. RS = first R then S; R° converse; $\subset$ containment; $\mathrm{Sub}(A)$ subobjects; $f^\#$ inverse image; $\mathrm{Dom}\, R = 1 \cap RR^\circ$; $\mathcal{S}$ = sets.

§1. Category tower (§1.4–1.9)

cartesian	finite products + equalizers (= all finite limits)
regular	cartesian + images; pullbacks transfer covers
pre-logos	regular; each $\mathrm{Sub}(A)$ a lattice, each $f^\# : \mathrm{Sub}(B) \to \mathrm{Sub}(A)$ a lattice map
logos	regular; each $\mathrm{Sub}(A)$ a lattice; $f^\#$ has a right adjoint: $f^\# B' \subset A' \iff B' \subset f^{\#\#} A'$
pre-topos	$\triangleq$ effective positive pre-logos
topos	cartesian; every object has a power-object ($\ni \subset [C] \times C$ universal). Thm: topos = positive effective transitive logos

§2. Category adjectives (§1.5–1.7)

well-supported A	$\mathrm{Spt}(A) \triangleq \mathrm{im}(A \to 1) = 1$
capital	every well-supported object covered by its points; then 1 is projective
positive	disjoint complemented $A \hookrightarrow C \hookleftarrow B \implies$ finite coproducts
effective	every equivalence relation is the level (kernel pair) of a map
boolean	every $\mathrm{Sub}(A)$ boolean; boolean pre-topos $\iff$ all diagonals complemented
AC	every entire relation contains a map $\iff$ every object projective
transitive	every endo-relation has a transitive closure $R^\dagger$

§3. Allegory tower (§2.1–2.4)

allegory	category $+R^\circ, \cap$: $(RS)^\circ = S^\circ R^\circ$, $R(S \cap T) \subset RS \cap RT$, modular law $RS \cap T \subset (R \cap TS^\circ)S$
distributive	$+0, \cup$: $R0 = 0$, $R(S \cup T) = RS \cup RT$, $R \cap (S \cup T) = (R \cap S) \cup (R \cap T)$
division	$+R/S$: $TS \subset R \iff T \subset R/S$
power	division $+\ni$: thick $1 \subset (R/\ni)(\ni/R)$, straight $(\ni/\ni) \cap (\ni/\ni)^\circ \subset 1$
derived	$\Lambda(R) \triangleq (R/\ni) \cap (\ni/R)^\circ$ is the unique map with $\Lambda(R)\ni = R$; power-object $[\alpha]$ = source of $\ni_\alpha$; $0, \cup$ become definable

§4. Allegory adjectives (§2.1–2.2)

unitary	has a unit λ: 1_λ its maximal endo, entire morphisms into λ from everywhere; λ = terminator of Map
tabular	every $R = f^\circ g$ with $ff^\circ \cap gg^\circ = 1$ (f, g maps) $\iff$ pre-tabular + coreflexives split
pre-tabular	every $R \subset$ some tabular morphism
effective	equivalence relations split ($E = ff^\circ$, $f^\circ f = 1$) $\iff$ Map effective
positive	distributive + finite coproducts ($\iff$ products; self-dual via $^\circ$)
semi-simple	every $R = F^\circ G$ with F, G simple $\iff$ splitting all symmetric idempotents is tabular
complete	locally: homs complete lattices, $R\big(\bigcup_i S_i\big) = \bigcup_i RS_i$; globally: + disjoint unions of object families. Locally complete distributive $\implies$ division: $R/S = \bigcup\{T : TS \subset R\}$

§5. The bridge (§1.56, §2.1)

$\mathrm{Rel}(C)$	objects of C; $(\alpha, \beta) = \mathrm{Sub}(\alpha \times \beta)$; composition = pullback-then-image; C regular $\implies$ associative + modular; $C \hookrightarrow \mathrm{Rel}(C)$ by graphs
$\mathrm{Map}(A)$	subcategory of maps = entire ($1 \subset RR^\circ$) + simple ($R^\circ R \subset 1$); on maps $\subset$ is =
roundtrip	$C \cong \mathrm{Map}(\mathrm{Rel}\, C)$ (C regular); $A \cong \mathrm{Rel}(\mathrm{Map}\, A)$ (A tabular)
regular data	pullback of f, g tabulates fg°; equalizer tabulates $\mathrm{Dom}(f \cap g)$; image of f tabulates $\mathrm{Dom}(f^\circ)$; g cover $\iff 1 \subset g^\circ g$
headline	small regular categories $\cong$ small unitary tabular allegories

§6. Dictionary (§2.1–2.4; A unitary tabular)

C is	**$\iff$ $\mathrm{Rel}(C)$ is**
regular	allegory
pre-logos	distributive
positive pre-logos	positive
logos	division
topos	power
effective	effective
pre-topos	distributive, positive, effective

§7. Reflections, completions (§2.16–2.5, §1.54)

All faithful representations; full where noted.

$\mathrm{Sid}(\mathcal{E})$	split symmetric idempotents $\mathcal{E}$; full + faithful when all identities $\in \mathcal{E}$
tabular refl.	$\mathrm{Sid}(\mathrm{Cor})$ of a pre-tabular allegory is tabular
effective refl.	$\mathrm{Sid}(\mathrm{Eq})$ of a tabular allegory; of a pre-power allegory: a **power** allegory
positive refl.	A^+ = finite matrices; full; preserves tabular, division
completions	ideals: any distributive allegory $\hookrightarrow$ locally complete; A^Σ (infinite matrices): globally complete; systemic $\triangleq \mathrm{Sid}(A^\Sigma)$ — of a small locally complete distributive allegory: a power allegory
pre-topos refl.	pre-logos $C \mapsto \mathrm{Map}(\mathrm{Sid}(\mathrm{Eq})((\mathrm{Rel}\, C)^+))$; same recipe: logos $\hookrightarrow$ positive effective logos (full)
amenable quot.	congruence respecting $\cup$, classes have maxima R^+; $\overline{R} \subset \overline{S} \iff R^+ \subset S^+$; preserves division, effective power, completeness
boolean quot.	identify R, S disjoint from the same things; of a tabular unitary division allegory: Map = a boolean logos; every topos $\hookrightarrow$ a boolean topos
capitalization	(category side) every small (pre-)regular category $\hookrightarrow$ a capital one

§8. Representation theorems (§1.55, §2.15–2.33)

Henkin–Lubkin	every small pre-regular category $\hookrightarrow$ a power of $\mathcal{S}$; so Horn sentences true in $\mathcal{S}$ hold in every regular category
allegories	every small unitary tabular allegory $\hookrightarrow$ a power of $\mathrm{Rel}(\mathcal{S})$ ($\triangleq$ representable)
distributive	small unitary distributive, pre-tabular **or** semi-simple $\hookrightarrow$ a power of $\mathrm{Rel}(\mathcal{S})$
division	tabular unitary division $\hookrightarrow$ $\mathcal{V}$-valued sets, $\mathcal{V}$ = opens of a Stone space; countable: $\hookrightarrow$ a countable power of $\mathcal{O}(X)$-valued sets, X metrizable without isolated points
logoi	countable logos $\hookrightarrow$ a countable power of $\mathrm{Sh}(X)$; with a coprime terminator: $\hookrightarrow \mathrm{Sh}(X)$

§9. Boundaries (§2.15, §2.4)

modular law	entire-or-empty relations satisfy every axiom **except** the modular law, and are tabular — the law is independent
Desargues	Veblen–Wedderburn plane: a one-object allegory (184 morphisms) representable in **no** power of $\mathrm{Rel}(\mathcal{S})$
no finite base	no finite equation set true in $\mathrm{Rel}(\mathcal{S})$ yields all equations of $\mathrm{Rel}(\mathcal{S})$; graph containments are decidable
effectivity	assemblies over partial recursive functions: a positive pre-logos, **not** effective; its effective reflection = the realizability topos